POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 4022 | Semester IV | Theory | 4 Credits

Machine Tools

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Mechanical Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 4022 as Machine Tools in Semester IV for Mechanical Engineering. The official SITTTR detailed source was unavailable during this cycle. With user approval, explanatory coverage uses reliable public diploma machine-tool curriculum sources and is not represented as recovered official SITTTR Course 4022 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders. This handbook is for study preparation; actual machine operation requires institutional safety rules, trained supervision, approved SOPs, guards and PPE.

Verified source note: The official detailed source was inaccessible. Public sources supporting cutting mechanics, lathe/milling/drilling/shaping/grinding and safety are linked in References.

COURSE DASHBOARD

What You Are Expected to Learn

60

Contact hours

4

Credits

Theory CIE 40

CIE

Theory ESE 60

ESE

Course introduction

Machine Tools introduces the principles, selection and safe operation of conventional machining equipment. It connects cutting mechanics, tool/work holding and process parameters with turning, milling, drilling, shaping and grinding operations used to produce accurate components.

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Prerequisite foundation

Engineering materials, manufacturing basics, workshop drawing/measurement and mandatory machine-shop safety discipline.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain metal-cutting mechanics, cutting parameters, tool materials and machining safety.
2. Describe lathe, milling and drilling machine construction, operations and work/tool holding.
3. Explain shaping, planning, grinding and finishing processes.
4. Select a suitable process and reasoned parameters for a simple component under competent workshop supervision.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

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Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain cutting tools, chip formation, cutting parameters and safe machining practice.	Understand
CO2	Select basic operations, work holding and process settings for lathe, milling and drilling tasks.	Apply
CO3	Explain shaping, planning, grinding and finishing-machine principles and applications.	Understand
CO4	Prepare a safe, documented operation sequence for an instructor-approved simple machining job.	Apply

Official CO-PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	Machining Fundamentals and Safety	Metal removal, chip formation, tool geometry/materials, cutting speed/feed/depth, MRR, cutting fluids, inspection and safety.	Approx. 14 hours	CO1	Module 1
2	Turning and Lathe Operations	Lathe parts/kinematics, work holding, turning/facing/threading/tapering, tools, settings, time and inspection.	Approx. 16 hours	CO2	Module 2
3	Milling and Hole-Making	Horizontal/vertical milling, cutters, indexing, up/down milling, drilling, reaming, boring and work/tool holding.	Approx. 16 hours	CO2	Module 3
4	Reciprocating and Abrasive Processes	Shaping/planning, quick return, grinding machines/wheels, dressing/truing, honing/lapping/superfinishing and process selection.	Approx. 14 hours	CO3	Module 4

LEARNING ROADMAP

How the Modules Connect

Machining Fundamentals and Safety

Metal removal, chip formation, tool geometry/materials, cutting speed/feed/depth, MRR, cutting fluids, inspection and safety.

CO1 · Approx. 14 hours

Turning and Lathe Operations

Lathe parts/kinematics, work holding, turning/facing/threading/tapering, tools, settings, time and inspection.

CO2 · Approx. 16 hours

Milling and Hole-Making

Horizontal/vertical milling, cutters, indexing, up/down milling, drilling, reaming, boring and work/tool holding.

CO2 · Approx. 16 hours

Reciprocating and Abrasive Processes

Shaping/planning, quick return, grinding machines/wheels, dressing/truing, honing/lapping/superfinishing and process selection.

CO3 · Approx. 14 hours

MODULE 1

Machining Fundamentals and Safety

Metal removal, chip formation, tool geometry/materials, cutting speed/feed/depth, MRR, cutting fluids, inspection and safety.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

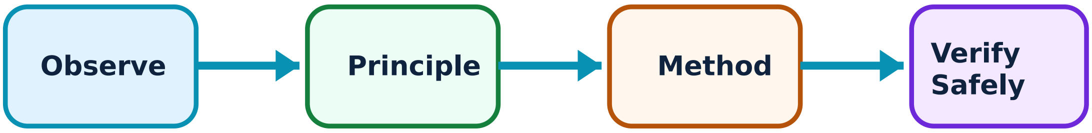
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Machining Fundamentals and Safety: identify the system, state its principle, follow the correct method and verify the result safely.

1. Metal cutting and chip formation−

Machining removes material by controlled relative motion between tool and workpiece. Tool geometry, work material, speed and feed influence chip form, forces, temperature and finish.

Study and application method

Identify work/tool/material removal motion, state cutting parameters with units and relate a parameter change to expected surface-finish, tool-life or productivity effect.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Identify work/tool/material removal motion, state cutting parameters with units and relate a parameter change to expected surface-finish, tool-life or productivity effect.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Never reach near a rotating part or chip; use approved chip-removal tools and machine guards.

2. Cutting parameters and process planning−

Cutting speed, feed and depth of cut affect material-removal rate, tool life, power and finish. Selection depends on machine, tool, material, rigidity and approved data.

Study and application method

Use instructor-approved tables, calculate in consistent units and write the setup, operation sequence, inspection points and safe stop conditions.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Use instructor-approved tables, calculate in consistent units and write the setup, operation sequence, inspection points and safe stop conditions.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not select speed/feed from memory or exceed machine/tool limits.

3. Cutting fluids and safety−

Cutting fluids may cool, lubricate and assist chip control; their selection and disposal follow material/process/environmental requirements. Safe machining relies on training, PPE, clamping and housekeeping.

Study and application method

Complete a pre-start check: guards, workholding, tool security, clearance, PPE, fluid and emergency-stop awareness.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Complete a pre-start check: guards, workholding, tool security, clearance, PPE, fluid and emergency-stop awareness.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Loose clothing, jewellery, gloves near rotating spindles and distracted operation are serious hazards.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Turning and Lathe Operations

Lathe parts/kinematics, work holding, turning/facing/threading/tapering, tools, settings, time and inspection.

Approx. 16 hours | CO2 | Understand · Apply

Learning goals

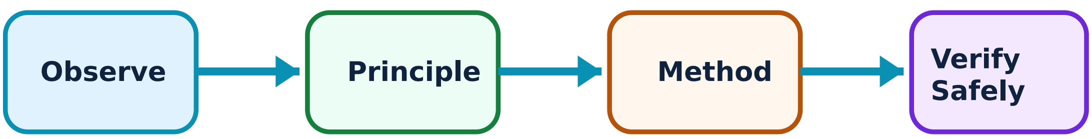
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Turning and Lathe Operations: identify the system, state its principle, follow the correct method and verify the result safely.

1. Lathe construction and work holding−

A lathe rotates the workpiece while a single-point tool feeds to generate cylindrical, flat or threaded features. Chucks, centres, faceplates, collets and mandrels suit different geometry and support needs.

Study and application method

Identify spindle, carriage, tailstock, feed system and selected workholding; explain how concentricity, support and clearance are ensured.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Identify spindle, carriage, tailstock, feed system and selected workholding; explain how concentricity, support and clearance are ensured.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Remove the chuck key immediately after tightening and verify clamping before starting.

2. Turning, facing, tapering and threading−

Lathe operations create diameters, faces, tapers, grooves and threads using controlled tool motion and gearing/attachments. Setup accuracy determines final geometry.

Study and application method

Read drawing dimensions/tolerances, create operation order from roughing to finishing, select tool/measurement method and record settings.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

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Common mistake / precaution: Stop the machine before measuring, adjusting or clearing chips.

3. Time, quality and troubleshooting−

Machining time and quality depend on travel, feed, speed, rigidity and tool condition. Chatter, poor finish, taper or dimensional error signal a setup/process issue.

Study and application method

Compare measured result with drawing, inspect tool/workholding/alignment and make one documented adjustment at a time.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare measured result with drawing, inspect tool/workholding/alignment and make one documented adjustment at a time.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not compensate for a fault by increasing speed or feed without identifying the cause.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Milling and Hole-Making

Horizontal/vertical milling, cutters, indexing, up/down milling, drilling, reaming, boring and work/tool holding.

Approx. 16 hours | CO2 | Understand · Apply

Learning goals

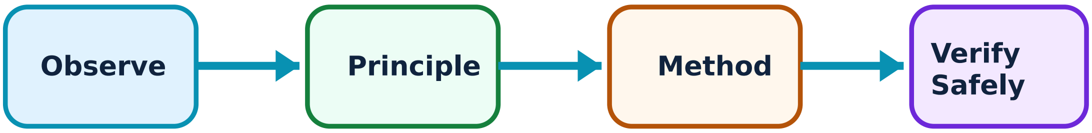
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Milling and Hole-Making: identify the system, state its principle, follow the correct method and verify the result safely.

1. Milling machines and cutters−

Milling uses a rotating multi-edge cutter while work is fed relative to it. Horizontal/vertical configurations, cutter type and workholding determine feasible features and finish.

Study and application method

Choose cutter, orientation, vice/fixture and feed direction from the required feature; identify possible climb/up-milling implications under the approved machine practice.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Choose cutter, orientation, vice/fixture and feed direction from the required feature; identify possible climb/up-milling implications under the approved machine practice.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Secure work and cutter, use correct spindle direction and never brush chips by hand near a rotating cutter.

2. Drilling, reaming and boring−

Drilling creates holes, reaming improves size/finish and boring enlarges/corrects an existing hole. Tool geometry, alignment, speed/feed and coolant influence accuracy.

Study and application method

Mark/locate as required, select drill/reamer/boring tool and workholding, plan pilot/finish sequence and inspect final diameter.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Mark/locate as required, select drill/reamer/boring tool and workholding, plan pilot/finish sequence and inspect final diameter.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Clamp workpiece; never hold it by hand while drilling.

3. Indexing and inspection−

Indexing divides rotation for equally spaced features. Measurement and in-process inspection control errors before they become scrap.

Study and application method

Calculate division with the approved indexing method, trial safely where instructed and use appropriate measuring tools with clean reference surfaces.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Calculate division with the approved indexing method, trial safely where instructed and use appropriate measuring tools with clean reference surfaces.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not change gears, belts or setup while power is on.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Reciprocating and Abrasive Processes

Shaping/planning, quick return, grinding machines/wheels, dressing/truing, honing/lapping/superfinishing and process selection.

Approx. 14 hours | CO3 | Understand · Apply

Learning goals

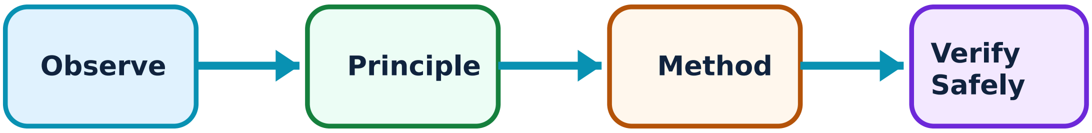
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Reciprocating and Abrasive Processes: identify the system, state its principle, follow the correct method and verify the result safely.

1. Shaping and planning−

Shapers and planers use reciprocating motion to produce flat or contoured surfaces. Quick-return mechanisms reduce idle stroke time; work/tool holding and stroke limits are essential.

Study and application method

Identify ram/table motion, cutting/return stroke, workholding and the feature to be produced; compare shaper versus planer by work/tool movement.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Identify ram/table motion, cutting/return stroke, workholding and the feature to be produced; compare shaper versus planer by work/tool movement.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Set stroke and clamping with machine stopped; keep clear of reciprocating travel.

2. Grinding and wheel selection−

Grinding removes small chips with an abrasive wheel for accuracy and finish. Abrasive, grain, grade, structure, bond, wheel speed and dressing affect performance.

Study and application method

Select a wheel only from approved data for work material/process, inspect/ring-test where policy requires, mount/guard correctly and dress as instructed.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Select a wheel only from approved data for work material/process, inspect/ring-test where policy requires, mount/guard correctly and dress as instructed.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: A damaged, wrongly mounted or over-speed wheel can fail catastrophically; follow all wheel-safety procedures.

3. Finishing and process choice−

Honing, lapping and superfinishing improve geometry, finish or bearing surface by controlled abrasive action. Process choice balances tolerance, finish, productivity, cost and material.

Study and application method

Compare required feature/tolerance/finish and propose a roughing-to-finishing sequence with inspection stages.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare required feature/tolerance/finish and propose a roughing-to-finishing sequence with inspection stages.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not claim a tolerance or finish without an appropriate measurement method and calibrated instrument.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

> For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

> For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

> For a comparison: use construction, working, result, advantage and limitation.

Recall 6

> For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Machining Fundamentals and Safety

Use the concept flow to revise observation, principle, method and verification.

Module 2: Turning and Lathe Operations

Use the concept flow to revise observation, principle, method and verification.

Module 3: Milling and Hole-Making

Use the concept flow to revise observation, principle, method and verification.

Module 4: Reciprocating and Abrasive Processes

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Prepare a safe operation plan and parameter sheet for an instructor-approved turning job.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Compare tool/workholding choices for a milled slot and a drilled/reamed hole.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Create a process sheet from drawing to inspection for a simple prismatic component.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Inspect a grinding-wheel setup only under approved workshop procedure and identify safety controls.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher’s approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Machining Fundamentals and Safety

Explain Metal cutting and chip formation and state one correct method, application or precaution.−

Machining removes material by controlled relative motion between tool and workpiece. Tool geometry, work material, speed and feed influence chip form, forces, temperature and finish.

Answer framework: Identify work/tool/material removal motion, state cutting parameters with units and relate a parameter change to expected surface-finish, tool-life or productivity effect.

Important point: Never reach near a rotating part or chip; use approved chip-removal tools and machine guards.

Explain Cutting parameters and process planning and state one correct method, application or precaution.−

Cutting speed, feed and depth of cut affect material-removal rate, tool life, power and finish. Selection depends on machine, tool, material, rigidity and approved data.

Answer framework: Use instructor-approved tables, calculate in consistent units and write the setup, operation sequence, inspection points and safe stop conditions.

Important point: Do not select speed/feed from memory or exceed machine/tool limits.

Explain Cutting fluids and safety and state one correct method, application or precaution.−

Cutting fluids may cool, lubricate and assist chip control; their selection and disposal follow material/process/environmental requirements. Safe machining relies on training, PPE, clamping and housekeeping.

Answer framework: Complete a pre-start check: guards, workholding, tool security, clearance, PPE, fluid and emergency-stop awareness.

Important point: Loose clothing, jewellery, gloves near rotating spindles and distracted operation are serious hazards.

Module 2 — Turning and Lathe Operations

Explain Lathe construction and work holding and state one correct method, application or precaution.−

A lathe rotates the workpiece while a single-point tool feeds to generate cylindrical, flat or threaded features. Chucks, centres, faceplates, collets and mandrels suit different geometry and support needs.

Answer framework: Identify spindle, carriage, tailstock, feed system and selected workholding; explain how concentricity, support and clearance are ensured.

Important point: Remove the chuck key immediately after tightening and verify clamping before starting.

Explain Turning, facing, tapering and threading and state one correct method,

application or precaution.−

Lathe operations create diameters, faces, tapers, grooves and threads using controlled tool motion and gearing/attachments. Setup accuracy determines final geometry.

Answer framework: Read drawing dimensions/tolerances, create operation order from roughing to finishing, select tool/measurement method and record settings.

Important point: Stop the machine before measuring, adjusting or clearing chips.

Explain Time, quality and troubleshooting and state one correct method, application or precaution.−

Machining time and quality depend on travel, feed, speed, rigidity and tool condition. Chatter, poor finish, taper or dimensional error signal a setup/process issue.

Answer framework: Compare measured result with drawing, inspect tool/workholding/alignment and make one documented adjustment at a time.

Important point: Do not compensate for a fault by increasing speed or feed without identifying the cause.

Module 3 — Milling and Hole-Making

Explain Milling machines and cutters and state one correct method, application or precaution.−

Milling uses a rotating multi-edge cutter while work is fed relative to it. Horizontal/vertical configurations, cutter type and workholding determine feasible features and finish.

Answer framework: Choose cutter, orientation, vice/fixture and feed direction from the required feature; identify possible climb/up-milling implications under the approved machine practice.

Important point: Secure work and cutter, use correct spindle direction and never brush chips by hand near a rotating cutter.

Explain Drilling, reaming and boring and state one correct method, application or precaution.−

Drilling creates holes, reaming improves size/finish and boring enlarges/corrects an existing hole. Tool geometry, alignment, speed/feed and coolant influence accuracy.

Answer framework: Mark/locate as required, select drill/reamer/boring tool and workholding, plan pilot/finish sequence and inspect final diameter.

Important point: Clamp workpiece; never hold it by hand while drilling.

Explain Indexing and inspection and state one correct method, application or precaution.−

Indexing divides rotation for equally spaced features. Measurement and in-process inspection control errors before they become scrap.

Answer framework: Calculate division with the approved indexing method, trial safely where instructed and use appropriate measuring tools with clean reference surfaces.

Important point: Do not change gears, belts or setup while power is on.

Module 4 — Reciprocating and Abrasive Processes

Explain Shaping and planning and state one correct method, application or precaution.−

Shapers and planers use reciprocating motion to produce flat or contoured surfaces. Quick-return mechanisms reduce idle stroke time; work/tool holding and stroke limits are essential.

Answer framework: Identify ram/table motion, cutting/return stroke, workholding and the feature to be produced; compare shaper versus planer by work/tool movement.

Important point: Set stroke and clamping with machine stopped; keep clear of reciprocating travel.

Explain Grinding and wheel selection and state one correct method, application or precaution.−

Grinding removes small chips with an abrasive wheel for accuracy and finish. Abrasive, grain, grade, structure, bond, wheel speed and dressing affect performance.

Answer framework: Select a wheel only from approved data for work material/process, inspect/ring-test where policy requires, mount/guard correctly and dress as instructed.

Important point: A damaged, wrongly mounted or over-speed wheel can fail catastrophically; follow all wheel-safety procedures.

Explain Finishing and process choice and state one correct method, application or precaution.−

Honing, lapping and superfinishing improve geometry, finish or bearing surface by controlled abrasive action. Process choice balances tolerance, finish, productivity, cost and material.

Answer framework: Compare required feature/tolerance/finish and propose a roughing-to-finishing sequence with inspection stages.

Important point: Do not claim a tolerance or finish without an appropriate measurement method and calibrated instrument.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Machining Fundamentals and Safety.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Turning and Lathe Operations.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Milling and Hole-Making.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Reciprocating and Abrasive Processes.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Metal removal, chip formation, tool geometry/materials, cutting speed/feed/depth, MRR, cutting fluids, inspection and safety..
2. Develop a complete answer or practical plan for: Lathe parts/kinematics, work holding, turning/facing/threading/tapering, tools, settings, time and inspection..
3. Develop a complete answer or practical plan for: Horizontal/vertical milling, cutters, indexing, up/down milling, drilling, reaming, boring and work/tool holding..

4. Develop a complete answer or practical plan for: Shaping/planning, quick return, grinding machines/wheels, dressing/truing, honing/lapping/superfinishing and process selection..

LAST-MILE REVISION

Quick Revision

Module 1: Machining Fundamentals and Safety

Metal removal, chip formation, tool geometry/materials, cutting speed/feed/depth, MRR, cutting fluids, inspection and safety.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Turning and Lathe Operations

Lathe parts/kinematics, work holding, turning/facing/threading/tapering, tools, settings, time and inspection.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Milling and Hole-Making

Horizontal/vertical milling, cutters, indexing, up/down milling, drilling, reaming, boring and work/tool holding.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Reciprocating and Abrasive Processes

Shaping/planning, quick return, grinding machines/wheels, dressing/truing, honing/lapping/superfinishing and process selection.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Metal cutting and chip formation	Machining removes material by controlled relative motion between tool and workpiece.
Cutting parameters and process planning	Cutting speed, feed and depth of cut affect material-removal rate, tool life, power and finish.
Cutting fluids and safety	Cutting fluids may cool, lubricate and assist chip control; their selection and disposal follow material/process/environmental requirements.
Lathe construction and work holding	A lathe rotates the workpiece while a single-point tool feeds to generate cylindrical, flat or threaded features.
Turning, facing, tapering and threading	Lathe operations create diameters, faces, tapers, grooves and threads using controlled tool motion and gearing/attachments.
Time, quality and troubleshooting	Machining time and quality depend on travel, feed, speed, rigidity and tool condition.
Milling machines and cutters	Milling uses a rotating multi-edge cutter while work is fed relative to it.
Drilling, reaming and boring	Drilling creates holes, reaming improves size/finish and boring enlarges/corrects an existing hole.
Indexing and inspection	Indexing divides rotation for equally spaced features.
Shaping and planning	Shapers and planers use reciprocating motion to produce flat or contoured surfaces.
Grinding and wheel selection	Grinding removes small chips with an abrasive wheel for accuracy and finish.
Finishing and process choice	Honing, lapping and superfinishing improve geometry, finish or bearing surface by controlled abrasive action.

LEARNING RESOURCES

References

Recommended machine-tools references

1. P. C. Sharma, Production Technology.
2. R. K. Jain and S. C. Gupta, Production Technology.
3. Hajra Choudhury, Bose and Roy, Elements of Workshop Technology.

Approved public machine-tool sources

- https://lit.laxmi.edu.in/psoacmee/2026/06/DI04000241.pdf
- https://www.ntc.edu/academics-training/programs/all/technical-diploma/machine-tool-technics/courses

These sources support the study handbook while official Course 4022 content is unavailable; they do not replace official syllabus requirements or workshop safety procedures.